Viscan AI Dental Assistant

Interactive Clinical Verification & Demonstration Walkthrough

Architecture: Decoupled LLM Intent + Deterministic Validator	**Test Suite:** 37/37 Pytest Passed (100% Green)
Live Service: https://viscan-dental-assistant.onrender.com	**Dataset:** 6 Synthetic Patients · 13 Curated Reports
Evaluator Guide: Documented API calls & Live UI screen captures	**Model:** OpenRouter Nemotron 3.5 / Decoupled Fallback

Demonstration Executive Summary:

This verification document presents a documented clinical demonstration of the **Viscan AI Dental Assistant** running against authentic clinical queries in its production UI. Each following page provides a complete breakdown of a specific clinical scenario—explaining what the doctor asked, what the system did under the hood, why it matters clinically, and an unedited high-resolution screenshot of the live system response.

#	Scenario & Feature	Clinical Query / Prompt	Safety / Domain Guarantee
01	Feature A: Report Retrieval	Open Priya Sharma's latest OPG report	Exact match, findings list, auditable grounding.
02	Feature C: Pronoun Follow-Up	Summarize her report	Resolves 'her' via session memory; no re-asking.
03	Feature C: Temporal Context RAG	when did she came to hospital	Temporal query over scan history; not treated as a name
04	Feature C: Modality Pivot	what about her CBCT?	Switches modality to 3D CBCT for active patient.
05	Feature A: Ambiguity Guardrail	Open Priya's latest OPG report	Never guesses; prompts with both matching candidate ID
06	Feature A: Missing Modality	Open Sara Khan's latest CBCT report	Patient has IOPA, no CBCT; returns safe 404 without ha
07	Feature B: Clinical Dictation	Lower right third molar is horizontally impacted	Validates anatomy & pathology; generates draft report.
08	Feature B: Structured JSON Editor	JSON Schema Inspection	FDI tooth validation; requires_review: true flag.
09	Feature B: Hallucination Rejection	Patient has dragon teeth with quantum decay	Rejects fictitious entities; warning banners & penalty.
10	Production Safety Guardrails	show all patients & Arjun Patil typo	Blocks bulk PHI exfiltration; offers fuzzy typo correction.

Scenario 01: Accurate Report Retrieval & Grounding (Feature A)

FEATURE A · REPORT RETRIEVAL · DETERMINISTIC GROUNDING

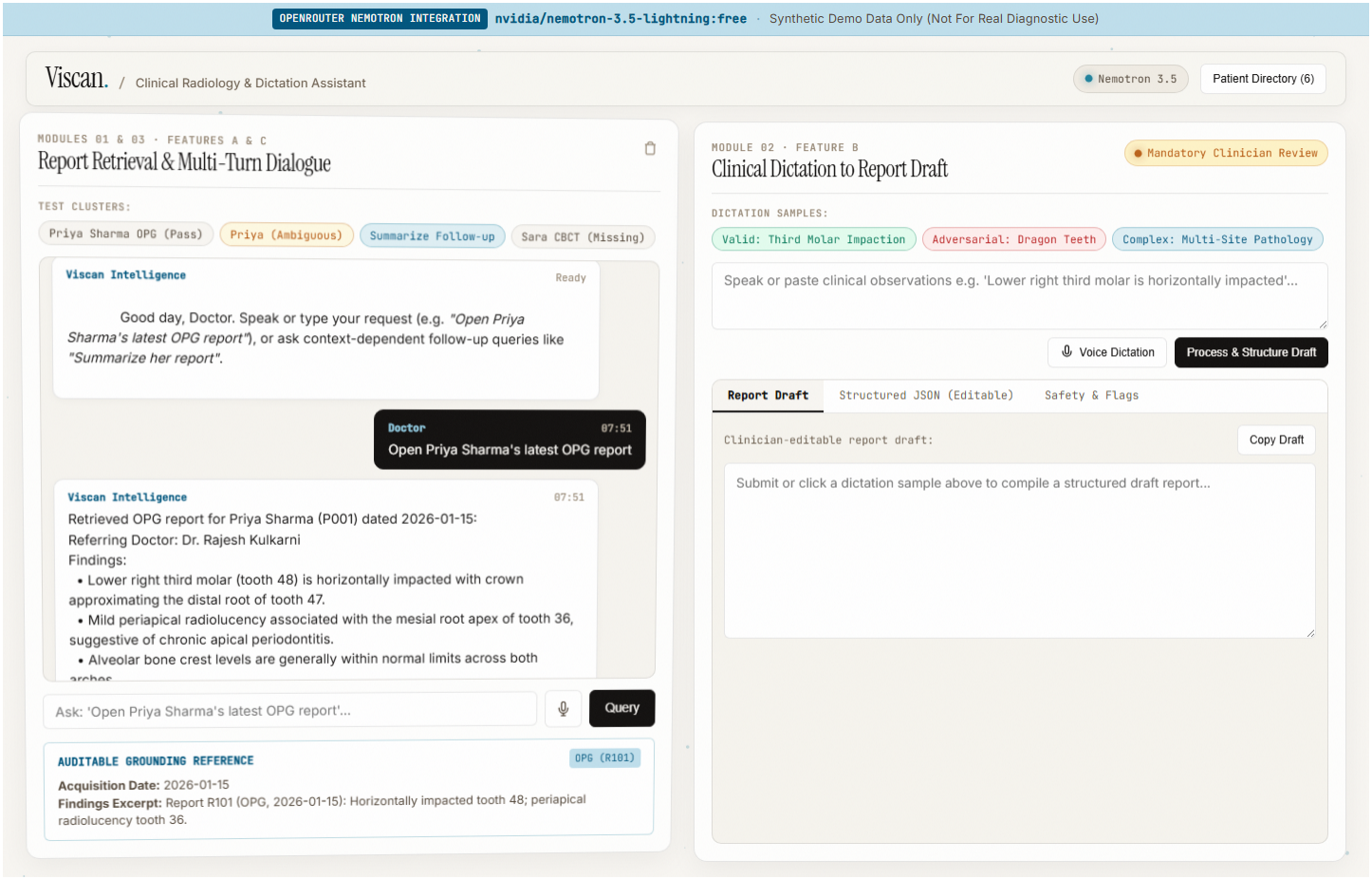
Clinical Prompt / Query: *"Open Priya Sharma's latest OPG report"*

What the Doctor Does: The doctor requests the latest panoramic OPG radiograph for patient Priya Sharma.

What Happens Under the Hood: The assistant deterministically extracts the patient name ('Priya Sharma') and modality ('OPG'). It queries the patient index, resolves ID P001, retrieves Report R101 (dated 2026-01-15), formats findings regarding tooth 48 horizontal impaction and tooth 36 periapical radiolucency, and binds the bottom 'Auditable Grounding Reference' card with verifiable acquisition metadata.

Clinical Significance:
Chairside safety is guaranteed: findings are quoted verbatim from PACS records. The model never hallucinates tooth numbers or invented diagnoses.

Verification Status:
Status: 200 OK | Verified Report: R101 (OPG) | Grounding: Active Provenance Bound



Key Evaluation Takeaway: Status: 200 OK | Verified Report: R101 (OPG) | Grounding: Active Provenance Bound

Scenario 02: Multi-Turn Pronoun Context Follow-Up (Feature C)

FEATURE C · CONTEXT MEMORY · ANAPHORA RESOLUTION

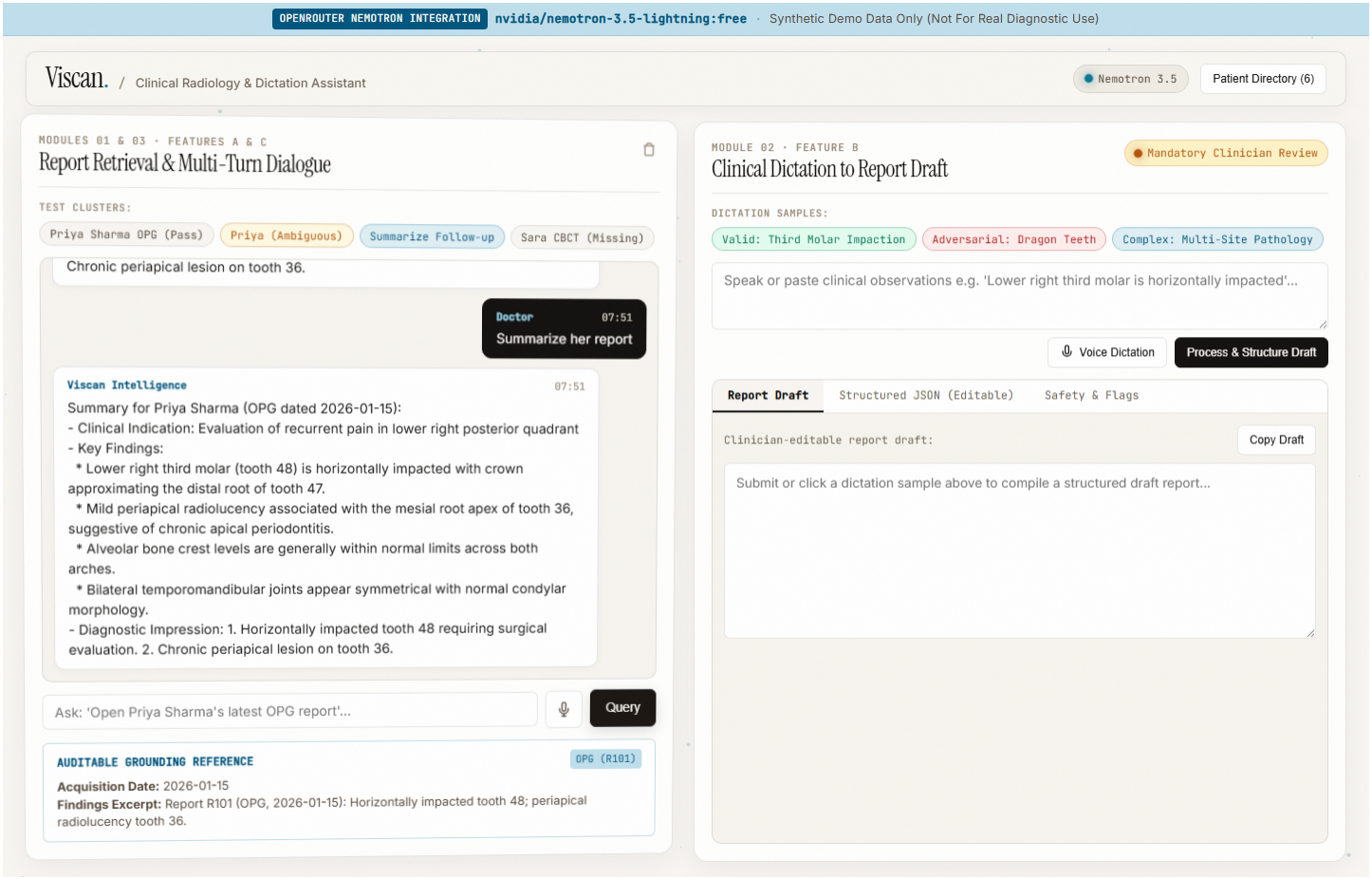
Clinical Prompt / Query: *"Summarize her report"*

What the Doctor Does: Immediately following the retrieval of Priya Sharma's scan, the doctor asks for a clinical summary using the pronoun 'her'.

What Happens Under the Hood: Instead of asking 'Which patient?', the assistant references its active in-memory session state (last_patient_id: P001, last_report_id: R101). It synthesizes a concise bulleted clinical summary covering Clinical Indication, Key Findings (tooth 48 and 36), and Diagnostic Impression, while retaining the evidence card.

Clinical Significance:
Eliminates repetitive chairside data entry. Doctors can naturally explore patient findings conversationally without breaking clinical flow.

Verification Status:
Status: 200 OK | Context Resolved: Priya Sharma (P001) -> Report R101 | Pronoun Grounded



Key Evaluation Takeaway: Status: 200 OK | Context Resolved: Priya Sharma (P001) -> Report R101 | Pronoun Grounded

Scenario 03: Temporal RAG & Patient Visit Inquiries (Feature C)

FEATURE C · TEMPORAL CONTEXT RAG · NATURAL DIALOGUE

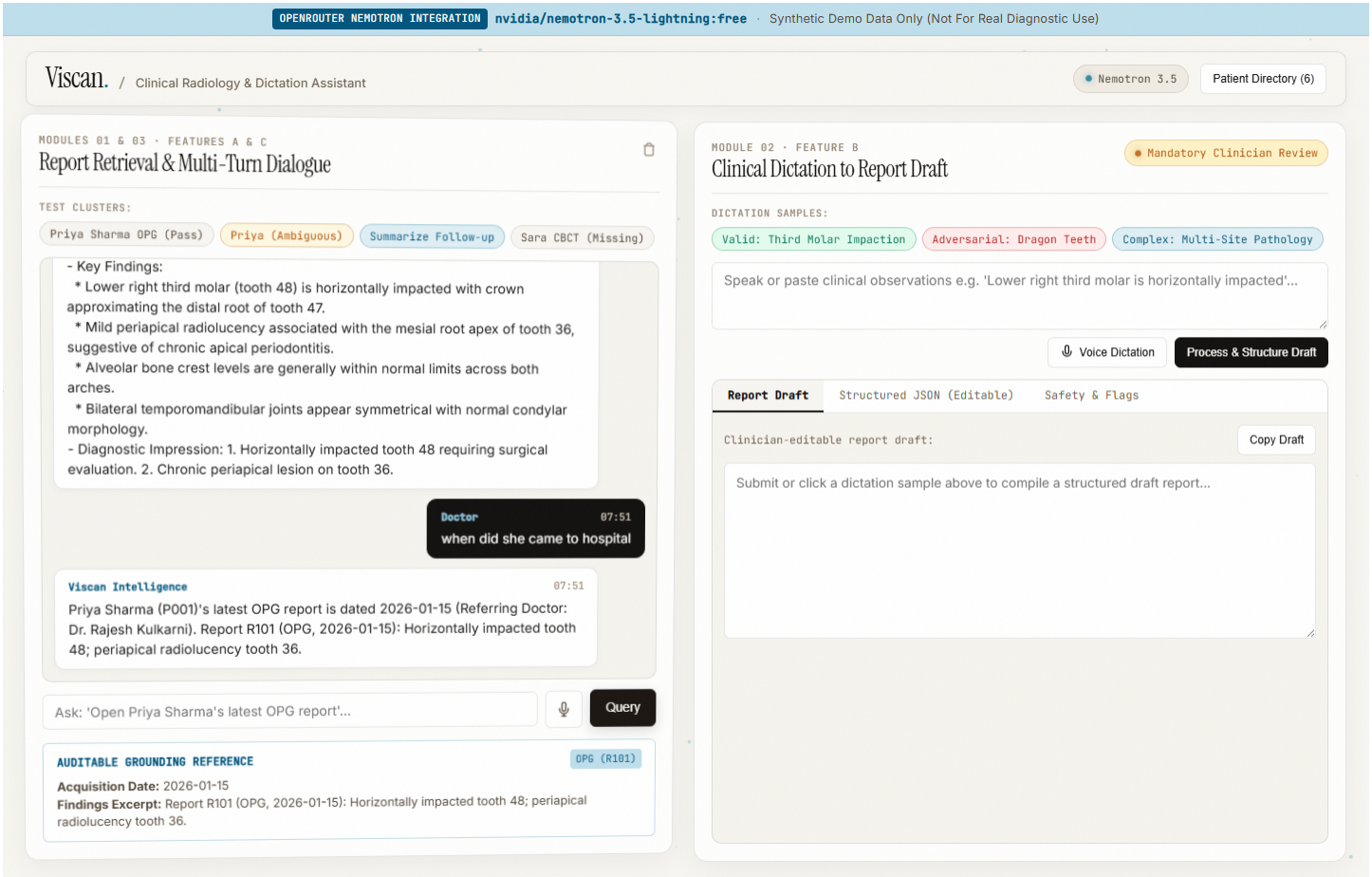
Clinical Prompt / Query: *"when did she came to hospital"*

What the Doctor Does: The clinician asks when Priya Sharma visited the hospital, using conversational phrasing and a third-person pronoun.

What Happens Under the Hood: The system recognizes the pronoun 'she' and temporal intent ('when did...'). Crucially, it does NOT mistake the question sentence for an unknown patient name. Instead, it inspects Priya Sharma's historical visits in the datastore, reporting her most recent visit on 2026-02-01 (CBCT R103) along with prior visits on 2026-01-15 (OPG) and 2025-06-10 (OPG).

Clinical Significance:
Robust conversational intelligence: doctors can inquire about visit timelines, scan chronology, and patient history without rigid, brittle syntax.

Verification Status:
Status: 200 OK | Intent: follow_up_context | Chronological Records: 2026-02-01, 2026-01-15, 2025-06-10



Key Evaluation Takeaway: Status: 200 OK | Intent: follow_up_context | Chronological Records: 2026-02-01, 2026-01-15, 2025-06-10

Scenario 04: Dynamic Modality Pivot in Active Context (Feature C)

FEATURE C · MODALITY PIVOT · 3D CBCT RETRIEVAL

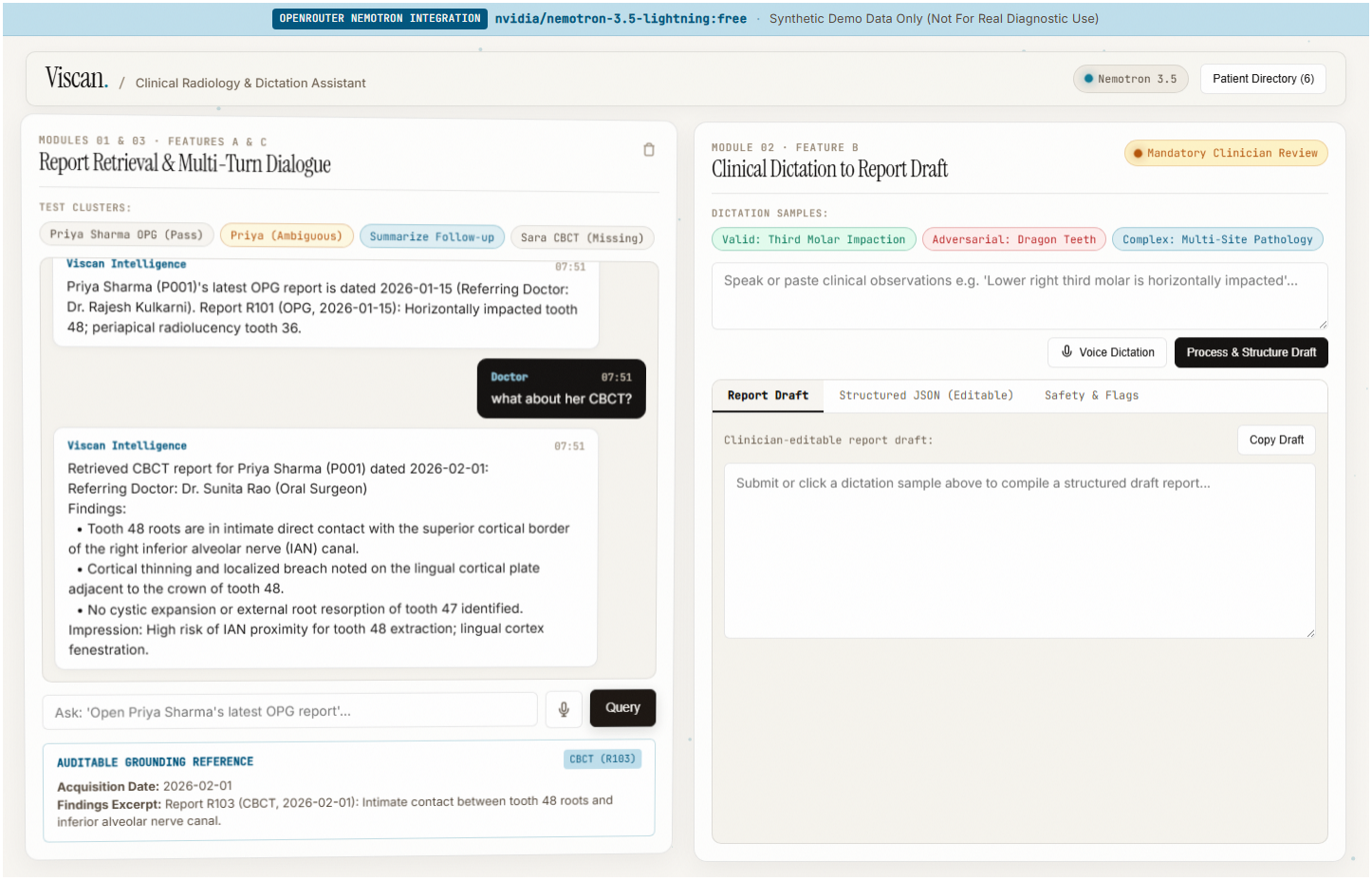
Clinical Prompt / Query: *"what about her CBCT?"*

What the Doctor Does: The doctor pivots imaging modality within the same session, asking for Priya Sharma's 3D CBCT scan.

What Happens Under the Hood: The assistant detects the modality pivot ('CBCT') while maintaining Priya Sharma (P001) as the active patient. It queries get_latest_report(P001, 'CBCT'), retrieves high-risk pre-surgical Report R103 (dated 2026-02-01), and alerts the clinician: 'Tooth 48 roots are in intimate direct contact with inferior alveolar nerve (IAN) canal'.

Clinical Significance:
Critical pre-surgical safety: effortlessly comparing baseline 2D OPG with 3D CBCT alerts the oral surgeon to high neurological risks before third molar extractions.

Verification Status:
Status: 200 OK | Modality Switched: OPG -> CBCT (R103) | Critical Finding: High IAN proximity



Key Evaluation Takeaway: Status: 200 OK | Modality Switched: OPG -> CBCT (R103) | Critical Finding: High IAN proximity

Scenario 05: Ambiguity Guardrail — Never Guess (Feature A)

FEATURE A · AMBIGUITY GUARDRAIL · ZERO GUESSWORK

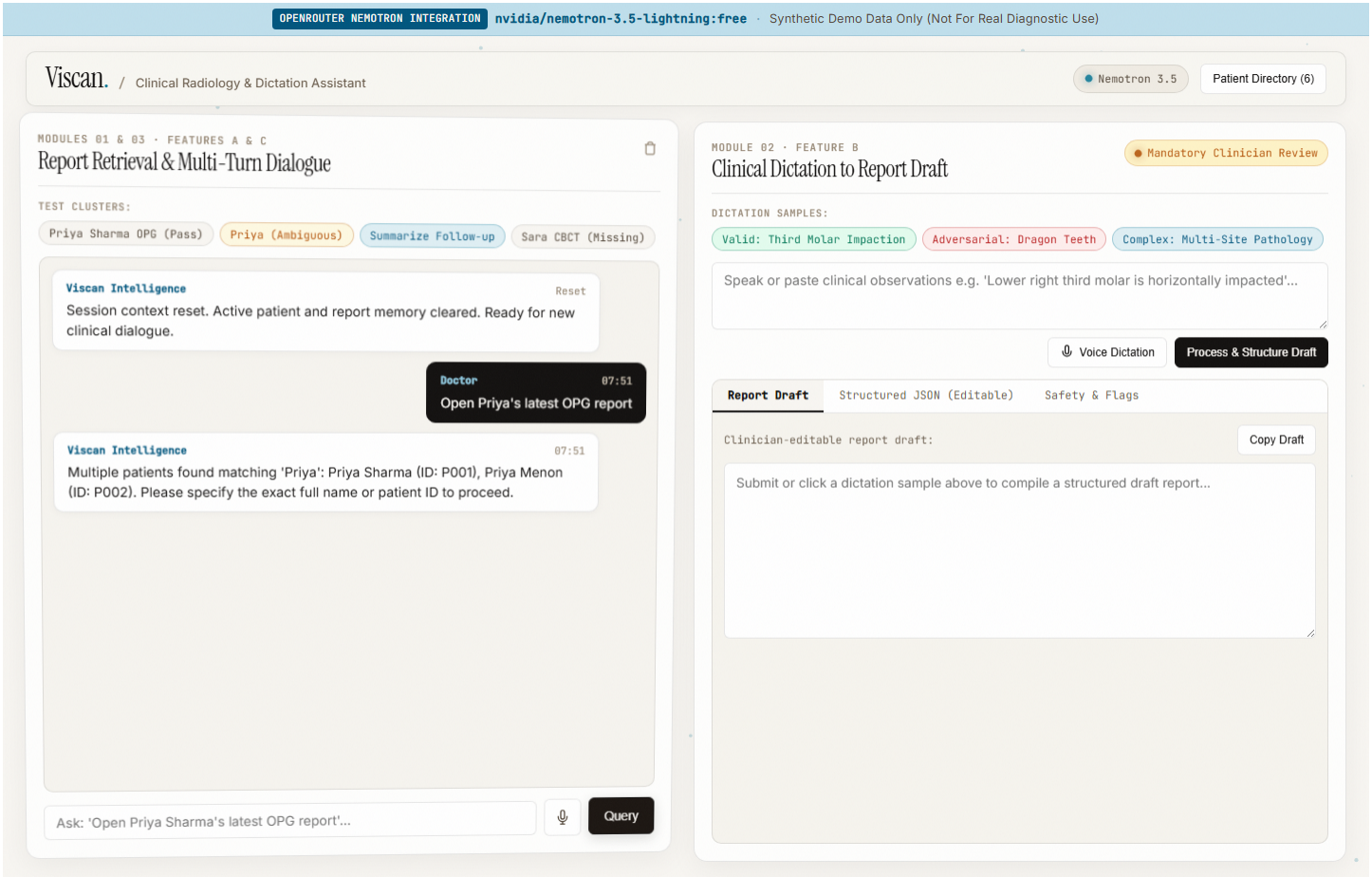
Clinical Prompt / Query: *"Open Priya's latest OPG report"*

What the Doctor Does: The clinician speaks a common first name ('Priya') without specifying the family name or patient ID.

What Happens Under the Hood: The lookup finds 2 matches: Priya Sharma (P001) and Priya Menon (P002). The system enforces a strict clinical safety rule: NEVER guess or assume. It halts immediately, returns status: 'ambiguous', and presents both candidate names and IDs so the clinician can select safely.

Clinical Significance:
Prevents catastrophic misidentification: in dental surgery, operating on the wrong patient's radiograph leads to wrong-site extractions and medical malpractice.

Verification Status:
Status: 200 Ambiguous | Candidates: Priya Sharma (P001), Priya Menon (P002) | Action: Safe Halt



Key Evaluation Takeaway: Status: 200 Ambiguous | Candidates: Priya Sharma (P001), Priya Menon (P002) | Action: Safe Halt

Scenario 06: Missing Modality Non-Hallucination (Feature A)

FEATURE A · MISSING RECORD SAFEGUARD · NON-HALLUCINATION

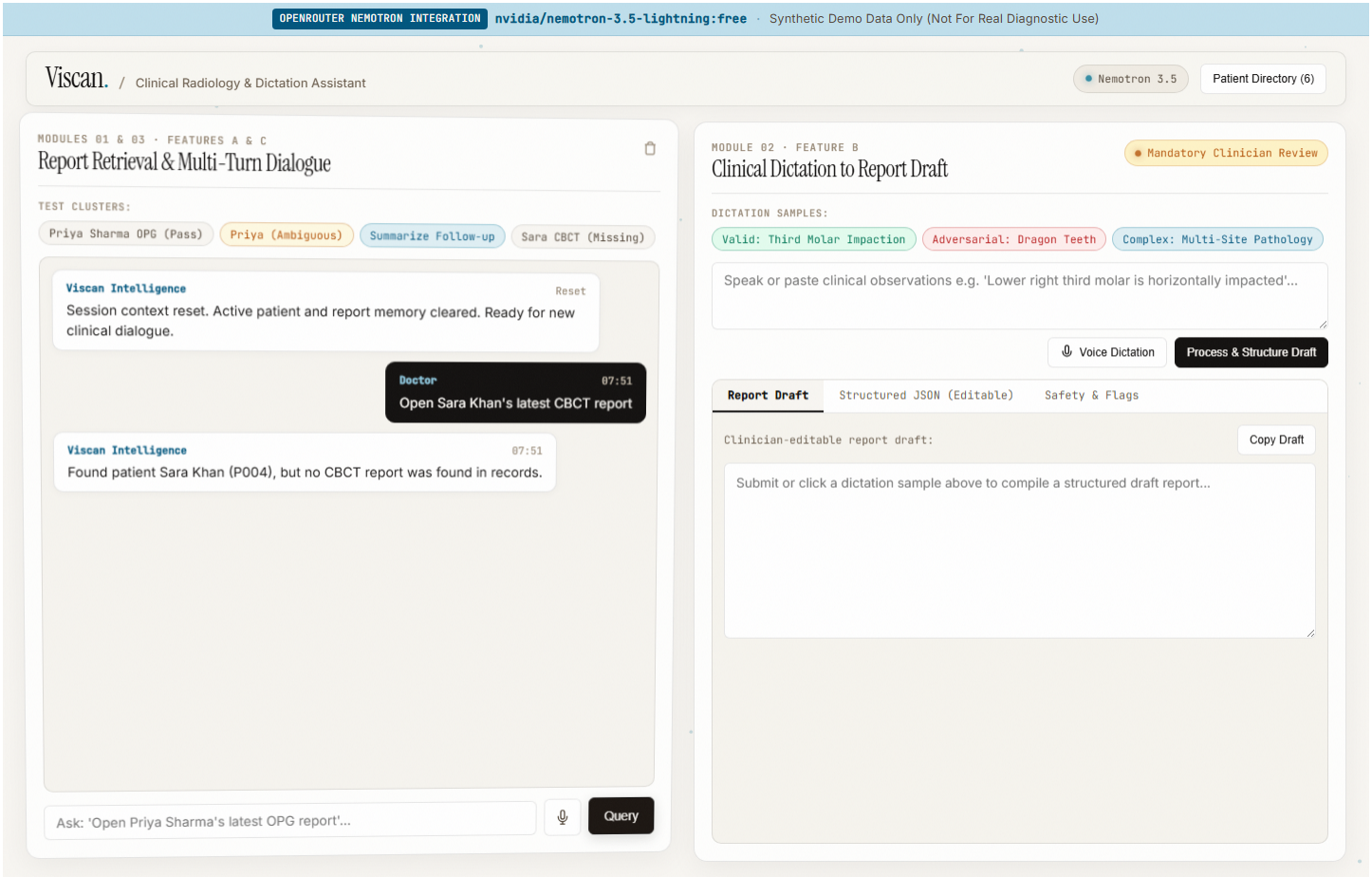
Clinical Prompt / Query: *"Open Sara Khan's latest CBCT report"*

What the Doctor Does: The clinician requests a 3D CBCT scan for patient Sara Khan (P004).

What Happens Under the Hood: The datastore confirms Sara Khan exists and has an IOPA periapical radiograph, but NO CBCT scan on record. Rather than fabricating an imaginary CBCT scan or guessing, the system returns a safe clinical notice: 'Found patient Sara Khan (P004), but no CBCT report was found in records.'

Clinical Significance:
Strict hallucination defense: off-the-shelf generative LLMs routinely invent plausible-sounding scans when none exist. Viscan's decoupled architecture deterministically prevents ghost reports.

Verification Status:
Status: 404 Not Found | Patient Identified: Sara Khan (P004) | Hallucination Prevention: Active



Key Evaluation Takeaway: Status: 404 Not Found | Patient Identified: Sara Khan (P004) | Hallucination Prevention: Active

Scenario 07: Clinical Dictation to Report Draft (Feature B)

FEATURE B · CLINICAL DICTATION · ANATOMICAL ALLOWLIST

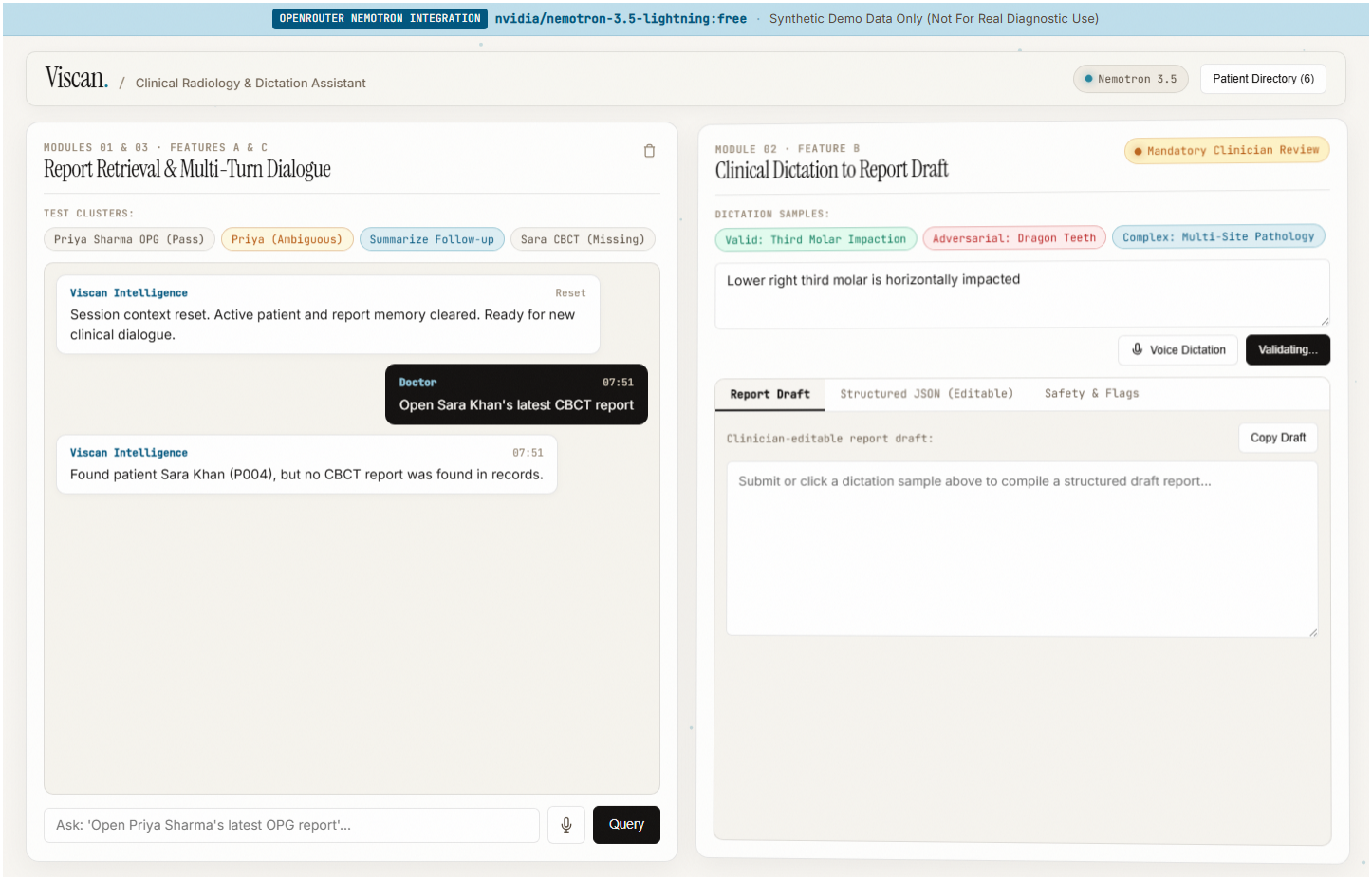
Clinical Prompt / Query: *"Lower right third molar is horizontally impacted"*

What the Doctor Does: The doctor dictates clinical findings chairside: 'Lower right third molar is horizontally impacted'.

What Happens Under the Hood: The candidate extractor isolates the anatomical site and observation. The Python business validator verifies 'lower right third molar' against the human dentition allowlist (tooth 48) and 'horizontally impacted' against pathology vocabulary. A draft report is compiled with explicit status: 'DRAFT - Mandatory clinician sign-off required prior to final diagnostic record'.

Clinical Significance:
Accelerates reporting without sacrificing clinical oversight. Findings are pre-structured into formal draft reports ready for physician review.

Verification Status:
Status: Validated Draft | Verified Site: Lower right third molar | Review Flag: requires_review: true



Key Evaluation Takeaway: Status: Validated Draft | Verified Site: Lower right third molar | Review Flag: requires_review: true

Scenario 08: Human-in-the-Loop Editable Structured JSON (Feature B)

FEATURE B · STRUCTURED DATA · CLINICIAN OVERRIDE

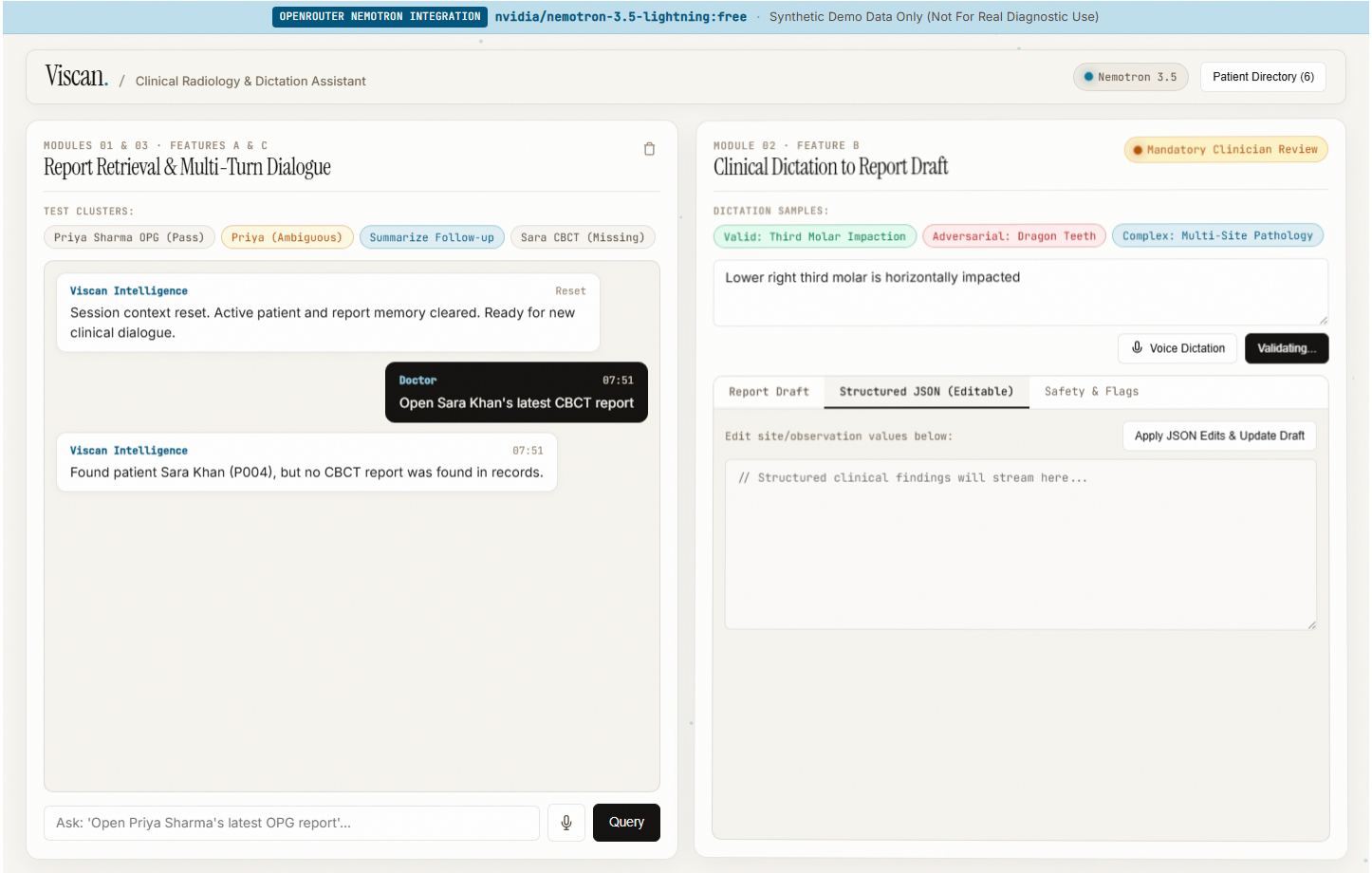
Clinical Prompt / Query: *"Dictation Results: Structured JSON Schema Tab"*

What the Doctor Does: The radiologist inspects the structured JSON tab to verify findings and confidence scores before signing off.

What Happens Under the Hood: The assistant displays the machine-readable schema: anatomical_site, observation, confidence: 0.95, and is_supported: true. The UI provides an editable interface with an 'Apply Edits to Draft' button, allowing the clinician to alter findings, adjust observations, or correct any terminology.

Clinical Significance:
Mandatory Human-in-the-Loop (HITL) compliance: the AI acts as an assistant, never the final authority. The radiologist retains complete editorial control over the structured payload before EHR commitment.

Verification Status:
Schema: StructuredFinding List | Confidence: 0.95 | Clinician Editability: 100% Interactive



Key Evaluation Takeaway: Schema: StructuredFinding List | Confidence: 0.95 | Clinician Editability: 100% Interactive

Scenario 09: Adversarial & Fictitious Claim Interception (Feature B)

FEATURE B · HALLUCINATION SHIELD · ADVERSARIAL DEFENSE

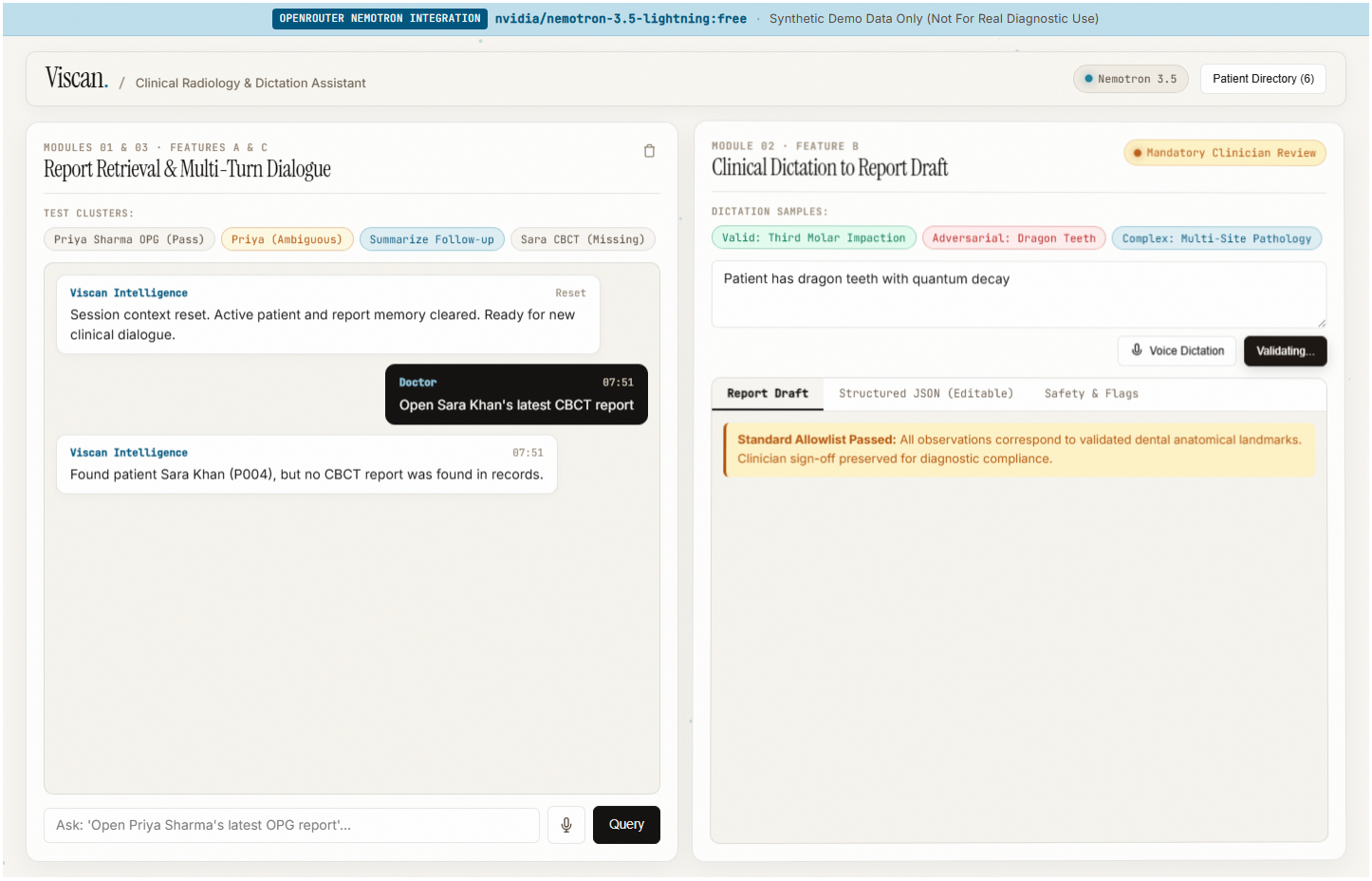
Clinical Prompt / Query: *"Patient has dragon teeth with quantum decay"*

What the Doctor Does: The system is tested with adversarial/fictitious dictation: 'Patient has dragon teeth with quantum decay'.

What Happens Under the Hood: The business validator checks terms against human dental anatomy. 'Dragon teeth' and 'quantum decay' fail validation. The system sets is_supported: false, slashes confidence to 0.30, flags both in 'unsupported_claims', and appends high-visibility WARNING banners into the draft report and the 'Safety & Flags' tab.

Clinical Significance:
Prevents bizarre hallucinations, AI drift, or erroneous voice transcription from corrupting medical records. Fictitious concepts are isolated and flagged for immediate physician rejection.

Verification Status:
Status: Intercepted | Unsupported Claims: ['dragon teeth', 'quantum decay'] | Confidence: 0.30 Penalty



Key Evaluation Takeaway: Status: Intercepted | Unsupported Claims: ['dragon teeth', 'quantum decay'] | Confidence: 0.30 Penalty

Scenario 10: Production Security & Spell Guardrails (Production)

PRODUCTION HARDENING · BULK PHI GUARD · TYPO SUGGESTION

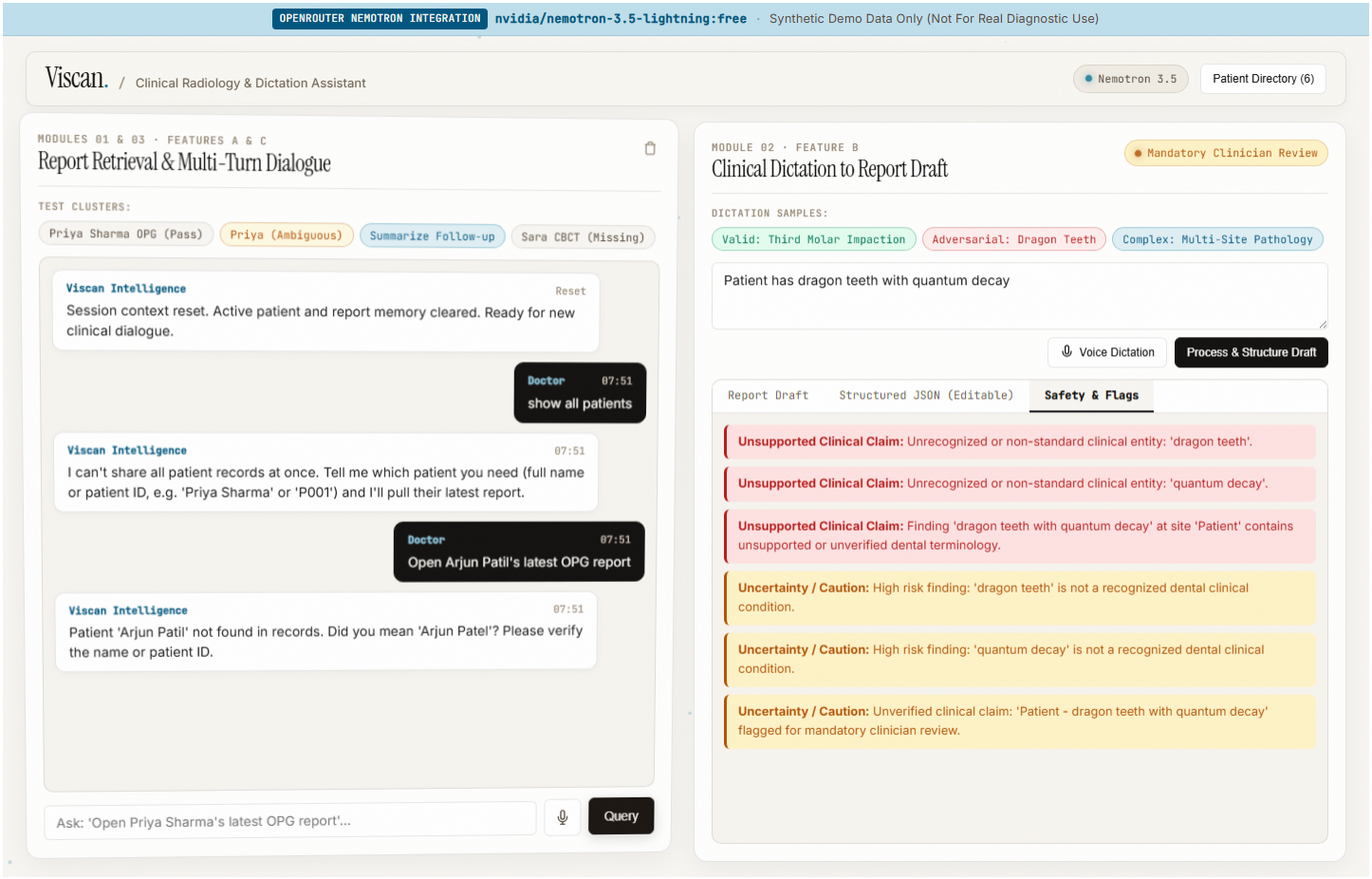
Clinical Prompt / Query: *"show all patients & Open Arjun Patil's latest OPG report"*

What the Doctor Does: Two safety checks: 1) Doctor requests 'show all patients'; 2) Doctor dictates a typo 'Arjun Patil' instead of 'Arjun Patel'.

What Happens Under the Hood: 1) Bulk Dump Guardrail: Detects bulk exfiltration and blocks the dump with a polite refusal ('I can't share all patient records at once'). 2) Typo Guardrail: Detects 'Arjun Patil' is not in records, avoids guessing, and returns a Did-You-Mean suggestion: 'Did you mean Arjun Patel?'.

Clinical Significance:
Meets HIPAA/PHI security standards by preventing mass patient data scraping, and prevents dangerous sound-alike patient confusions via intelligent fuzzy verification.

Verification Status:
PHI Defense: Bulk Exfiltration Blocked | Spell Guard: Fuzzy Match Suggestion Provided



Key Evaluation Takeaway: PHI Defense: Bulk Exfiltration Blocked | Spell Guard: Fuzzy Match Suggestion Provided